Bayesian Modeling of Rare Events in Financial Fraud Detection

Math 509:

Data Structures and Platforms

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1. Introduction

1.1: Background & Motivation

Credit card fraud is one of the most pervasive and costly problems in modern financial systems. While individual fraudulent transactions may seem isolated, their cumulative impact is enormous, both for the victims who lose money and for financial institutions that must absorb compensation costs. According to industry estimates, global card fraud losses exceed tens of billions of dollars annually.

What makes fraud detection uniquely challenging from a data science perspective is not the complexity of any single fraud, but rather its extreme rarity. In a typical credit card transaction dataset, fraudulent transactions make up less than 0.2% of all activity. This creates a deeply imbalanced problem: a model that simply labels every transaction as "legitimate" would achieve an accuracy of over 99.8% yet it would be completely useless at detecting actual fraud.

Additionally, the best model may not be the one that is best at predicting fraud but the one that leads to the least amount of money lost. While micro level threats can be mitigated, leading to an ideal world with no fraud, one must be aware of the “normal” limit or range of frauds to mitigate a macro level threat.

1.2: Problem Statement

The fundamental problem lies in not just finding a model that can predict fraudulent transactions, but generating the probability that a given transaction is fraud and the right threshold beyond which we need to take action. However, the goal is ultimately to protect the financial interests by prioritizing a model that leads to lower financial losses using a principled Bayesian statistical framework. This project aims to perform modelling of rare events and minimization of expected loss in credit card transactions through fraud detection prediction and prevention.

1.3: Project Objectives

Three specific, quantifiable questions are addressed in this project:

1. **Threshold Question:** What is the minimum probability score beyond which a transaction should be automatically flagged as fraudulent? Moreover, how many frauds does this catch versus how many legitimate customers does it inconvenience?
2. **Financial Loss Question:** Given that false negatives (missed frauds) cost the full transaction amount, and false positives (wrongly blocked cards) cost consumer time and company resources, what decision threshold minimises the bank's total financial damage?
3. **Systemic Monitoring Question:** Based on historical fraud rates, what is the expected daily range of fraudulent transactions? Exceeding or falling beyond this range is a signal which will require investigation.

1.4: Dataset Description

The dataset used in this project is the publicly available **European Credit Card Transactions Dataset**, containing transactions made by European cardholders over a two-day period in September 2013. It consists of:

- **284,807 total transactions**
- **492 confirmed fraud cases** (approximately 0.172% of all transactions)
- **30 feature columns:** Time, Amount, and 28 anonymised PCA-transformed features labeled V1 through V28
- **1 target column:** Class (0 = Normal, 1 = Fraud)

Because the raw features (V1–V28) were transformed via Principal Component Analysis (PCA) by the original data providers to protect cardholder privacy, we cannot directly interpret what each variable represents in human terms. However, the PCA transformation preserves the statistical structure of the original data, meaning the correlation patterns between these features and fraud status remain meaningful and explorable.
The two non-PCA features: Time (seconds elapsed since the first transaction) and Amount (transaction value in euros), were standardised using z-score normalisation prior to modelling.

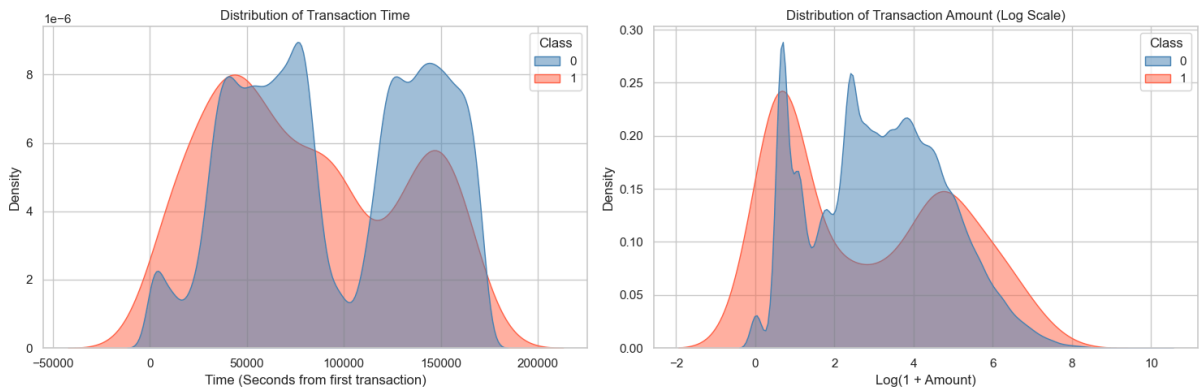


Figure 1 shows the distribution of transaction time and amount for normal (Class 0) vs. fraudulent (Class 1) transactions. Fraudulent transactions tend to cluster at lower amounts, though outliers at high amounts carry disproportionate financial risk.

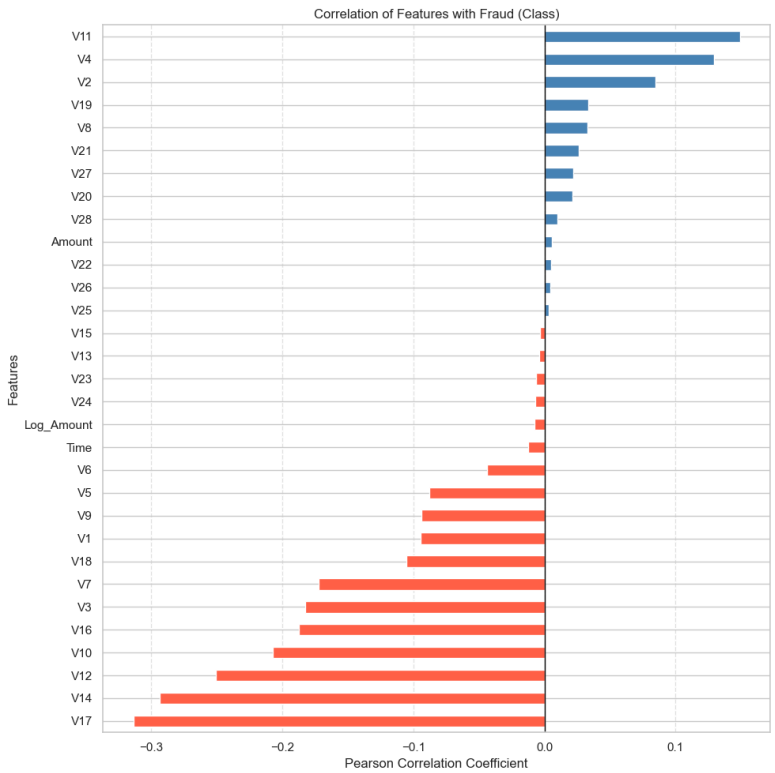
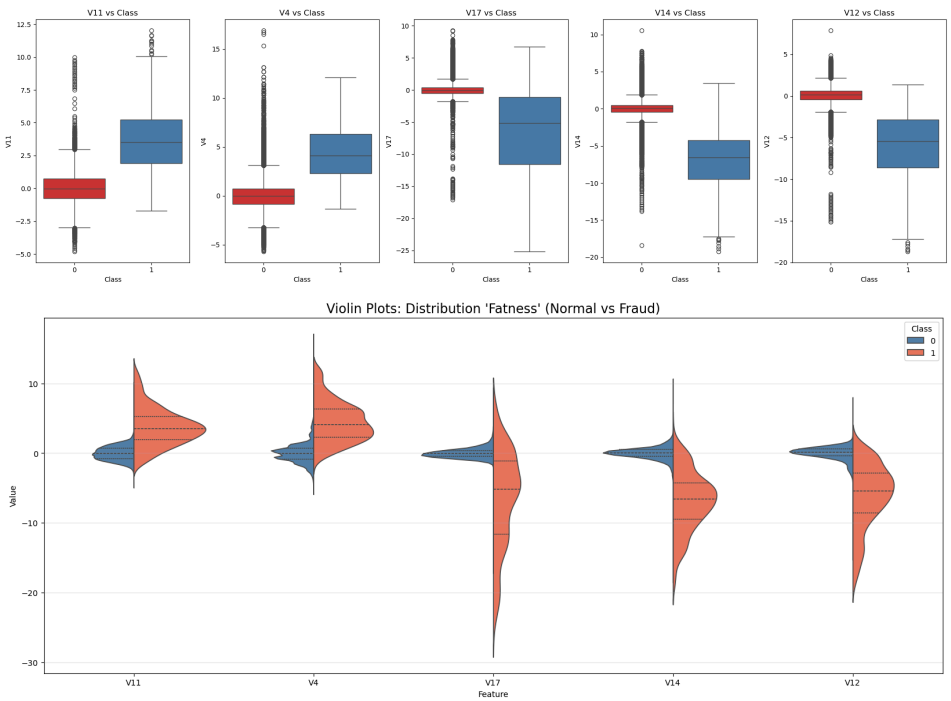


Figure 2 shows the correlation of each feature with the fraud label. The most discriminative features (V17, V14, V12, V10, and V16) were identified as the strongest negative correlators with fraud, while V4 and V11 showed the strongest positive correlation.



Figures 3 and 4 (box plots and violin plots) further confirm the separability of fraud vs. normal transactions across the top five features. The distributions for fraudulent transactions are distinctly different, with shifted medians and tighter spreads, indicating these features carry real signal.

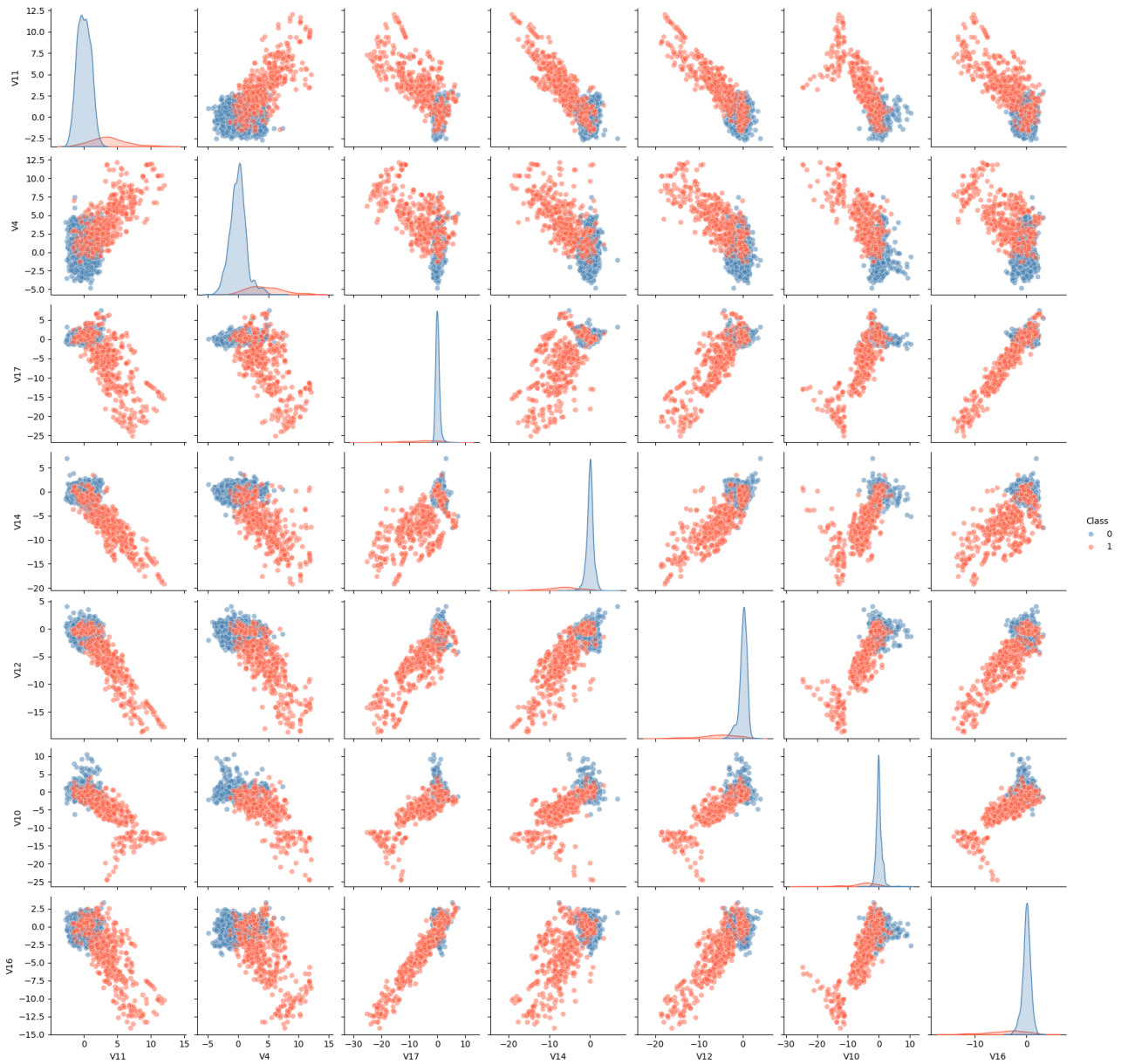


Figure 5 shows the pairwise distribution of the top seven candidate features. Low multicollinearity between features was confirmed through Variance Inflation Factor (VIF) analysis and all VIF scores were well below the critical threshold of 5, confirming independence.

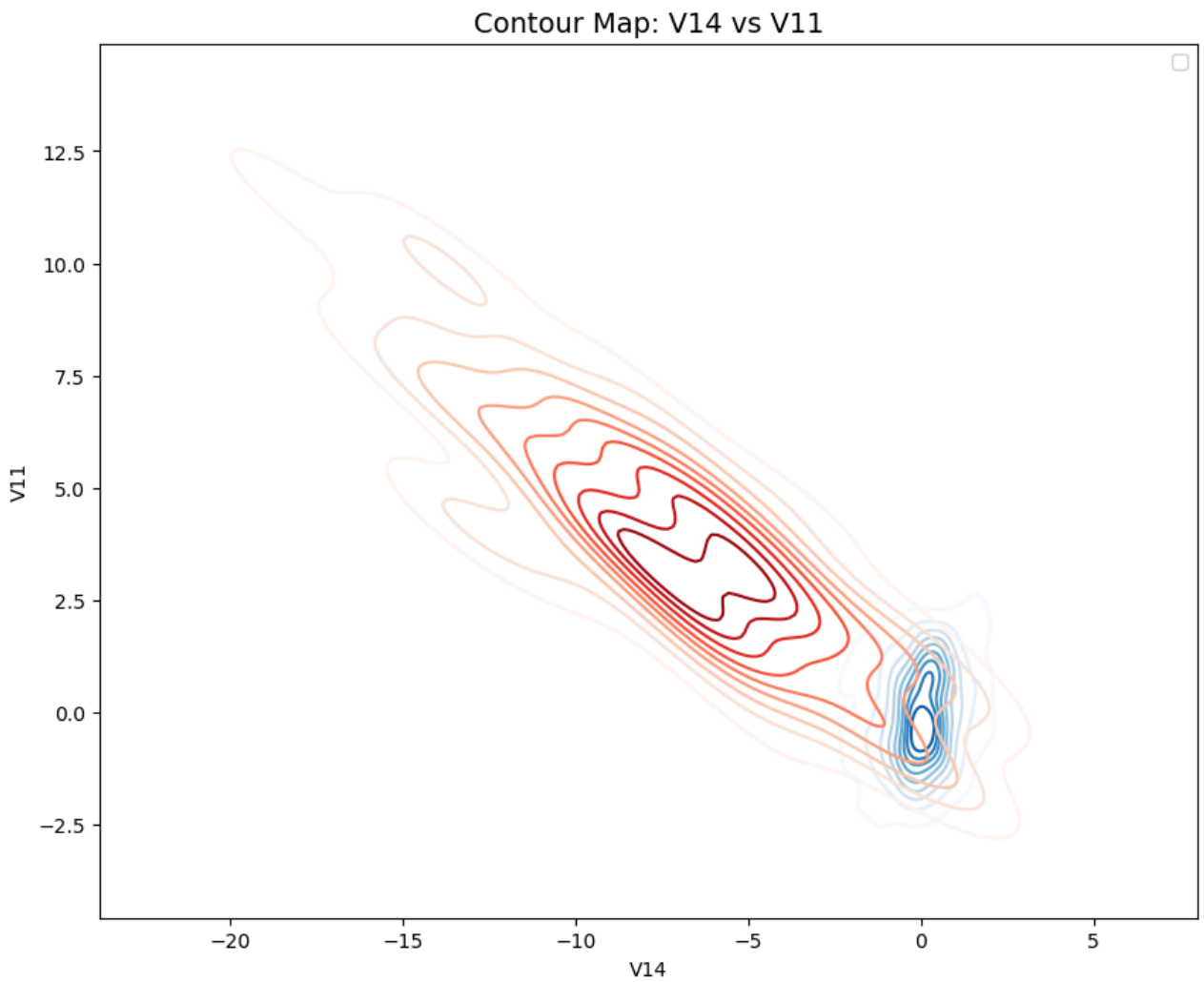


Figure 6 shows a contour plot of V14 vs.V11 for both classes. The near-complete separation of the red (fraud) and blue (normal) density regions visually confirms the feature's discriminating power.

2. Mathematical Model

2.1: Variables & Parameters

Symbol	Description	Value / Source
N	Total number of transactions	284,807
F_{obs}	Observed number of confirmed frauds	492
θ	Unknown true fraud probability	Estimated via MCMC
α, β	Beta distribution shape parameters (prior)	$\alpha = 1,\ \beta = 500$
x_i	Feature vector for transaction i	V17, V14, V12, V10, V16
x_k	Feature vector for transaction k (baseline ML)	V17, V14, V12, V4, V11
α_0	Intercept term in logistic model	Learned via MCMC
β	Coefficient vector for features	Learned via MCMC
$\sigma(\cdot)$	Sigmoid (logistic) function	$\sigma(z) = \frac{1}{1+e^{-z}}$
C_{FP}	Cost of a false positive (friction)	€10.00 (assumed)
C_{FN}	Cost of a false negative (compensation amount)	Variable - equals transaction amount
$\hat{\tau}$	Decision threshold for fraud classification	Optimised per objective
λ	Learning rate for gradient descent	Tuned adaptively

Table 1 shows the model variables and parameters.

2.2: Assumptions

1. **Conditional Independence:** Given the fraud label, individual transaction features are assumed to be approximately independent. VIF analysis confirmed this is reasonable.
2. **Stationarity:** The fraud mechanism is assumed to be consistent across the two-day observation window, which means that fraud patterns do not shift significantly over time.
3. **Reported = True:** The dataset labels a transaction as fraud only when a cardholder reported it. Unreported frauds are treated as normal transactions, this introduces a downward bias in the true fraud count.
4. **Friction cost is fixed:** For Objective 2, a false positive (incorrectly blocked card) incurs a fixed €10 administrative cost, independent of transaction amount.
5. **Full amount lost on false negative:** When a fraudulent transaction is missed (false negative), the bank compensates the cardholder for the full transaction amount.

6. **Imbalance requires resampling:** The extreme class imbalance (0.172% fraud rate) necessitates a balanced training set. The Bayesian model was trained on a 1:4 (fraud:normal) subsample to ensure adequate representation of both classes.
7. **MCMC convergence as quality criterion:** Model validity is assessed by the Gelman-Rubin statistic, i.e. values close to 1.0 indicate convergence and reliable posterior estimates.

2.3: Baseline ML Models

Before diving into the objectives, we first set a baseline with generic and widely adopted machine learning (ML) models. These models were: Logistic Regression, Random Forest, SVM (poly kernel), SVM (RBF kernel).

2.4: Objective 1: Fraud Prediction (Probability & Threshold Estimation)

The modelling proceeds in two stages: a simple global model for the marginal fraud rate, followed by a full Bayesian logistic regression for individual transaction classification.

2.4.1: Global Fraud Rate Model

The marginal fraud probability is modelled as:

$$F \mid \theta, N \sim Binomial(N, \theta)$$
$$\theta \sim Beta(\alpha = 1, \beta = 500)$$

The Beta(1, 500) prior encodes the belief that fraud is rare: a domain-informed prior representing roughly one fraud per 500 transactions. The posterior, updated with the 492 observed frauds out of 284,807 transactions, yields a tightly concentrated distribution around $\theta \approx 0.00173$.

2.4.2: Bayesian MCMC Sampler

For individual transaction classification, the model is:

$$P(Fraud_i = 1|x_i) = \sigma\left(\alpha_0 + \sum_{j=1}^{5} \beta_j \cdot x_{ij}\right) = \sigma(z_i)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function, and the features $\{x_{ij}\}$ are the five selected variables: V17, V14, V12, V10, V16.

Priors on coefficients are defined as skeptical (regularising) normals:

$$\beta_j \sim N(0, 0.1), \; j = 0, 1, \ldots, 5$$

Here, constrained prior $\sigma = 0.1$ prevents the MCMC sampler from exploring extreme coefficient values that would cause numerical instability, while still allowing the data to dominate in regions where evidence is strong.

2.5: Objective 2: Further Mitigation of Financial Loss

The total expected cost at threshold τ is:

$$L(\tau) = C_{FP} \cdot \sum_{i \in Normal} 1[\hat{p}_i \geq \tau] + \sum_{i \in Fraud} 1[\hat{p}_i < \tau] \cdot A_i$$

where A_i is the transaction amount for observation i. Gradient descent is applied to find the floor threshold $\hat{\tau}$ that minimises $L(\tau)$, with a stopping criterion to prevent the threshold from collapsing to zero (which would flag every transaction).

2.6: Objective 3: Confidence Interval for Daily Fraud Counts

Daily fraud volume is modelled as a Poisson process. The expected daily fraud count is:

$$\mu = N_{\text{daily}} \cdot \hat{\theta}$$

The 95% confidence interval is constructed from the Normal approximation to the Poisson distribution:

$$CI_{95\%} = \mu \pm 1.96\sqrt{\mu}$$

2.7: Directed Acyclic Graph (DAG)

The causal structure of the model is represented graphically. The fraud status (latent variable) acts as the root cause influencing all five feature variables, which in turn generate the observed transaction signatures.

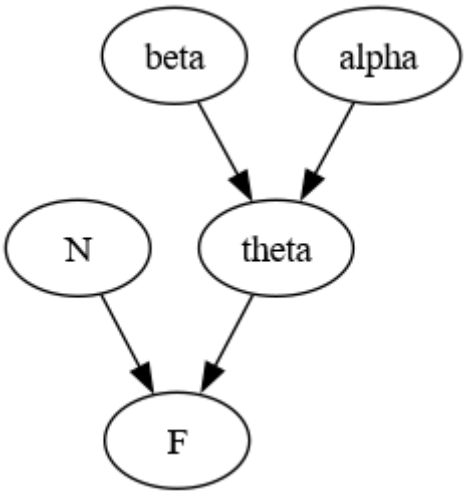


Figure 7 shows the directed acyclic graph of the model.

3. Solution

3.1: Data Preprocessing Pipeline

1. **Deduplication:** Duplicate rows were removed, reducing the dataset from 284,807 to 283,726 unique transactions.
2. **Feature Scaling:** Time and Amount were standardized using StandardScaler (zero mean, unit variance). The V1–V28 features were already PCA-scaled by the data provider.
3. **Feature Selection:** Pearson correlation between each feature and class was computed. The top 5 features by absolute correlation strength were retained for the Bayesian model:
   Note: All top 5 highest correlating variables are negatively correlated

Rank	Feature	Absolute Correlation
1	V17	0.3135
2	V14	0.2934
3	V12	0.2507
4	V10	0.2070
5	V16	0.1872

Table 2 shows the five selected features.

4. **Train/Test Split:** Stratified 80/20 split, preserving the original fraud ratio in both sets. This produced a training set of 226,980 rows (378 frauds) and a test set of 56,746 rows (95 frauds).
5. **Balanced Bayesian Training Set:** From the training data, all 378 fraud cases were retained, and 1,512 normal transactions were randomly sampled (1:4 ratio), yielding 1,890 training rows for the MCMC sampler.

3.2: Baseline Models

Three standard machine learning (ML) classifiers were trained on the full (unbalanced) training set as a performance baseline. These models do not incorporate uncertainty quantification and serve as a reference point.
Note: For Baseline ML models We used V17 V14 V12 V11 and V4 (here V4 and V11 are highest positively correlated variables) as they led to better results.

Model	Recall (Catch Rate)	Precision (Annoyance Rate)	F1-Score
Logistic Regression (Full Data)	77.89%	10.76%	0.19
Random Forest (Full Data)	***72.63%***	***97.18%***	***0.83***
SVM: RBF Kernel (5% Data)	30.53%	34.52%	0.32
SVM: Poly Kernel (5% Data)	62.11%	36.65%	0.46

Table 3 shows the model comparison of Precision, Accuracy and F1-Score.

Random Forest achieved the best overall F1-score of 0.83, making it the strongest baseline. However, it provides no probabilistic uncertainty quantification, and its threshold cannot be meaningfully adapted to financial cost structures.

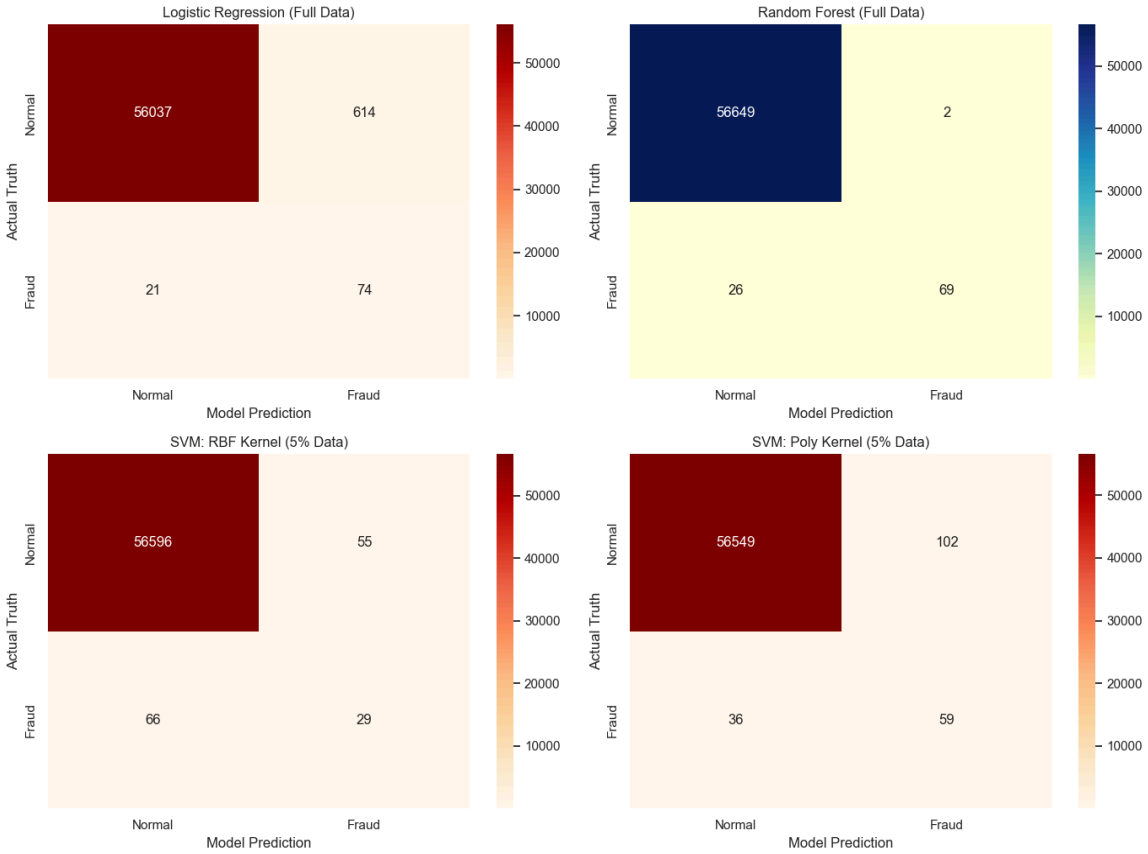


Figure 8 shows the confusion matrices of all 4 models (baseline).

3.3: MCMC Sampling (Objective 1)

The Bayesian logistic regression model was built using **PyMC** and sampled with the **No-U-Turn Sampler (NUTS)**, a variation of **Hamiltonian Monte Carlo (HMC)**. Configuration:

- **Chains:** 4 independent chains
- **Draws per chain:** 1,000
- **Tuning steps:** 1,000
- **Target acceptance rate**: 0.95
- **Random seed:** 42 (for reproducibility)
- **Jitter:** Small Gaussian noise ($N(0, 0.01)$) added to features to prevent numerical degeneracy

The MCMC traces for the 4-chain run are shown below. Healthy traces should resemble *"fuzzy caterpillars"*, which mean they are well-mixed chains that overlap completely, indicating convergence.

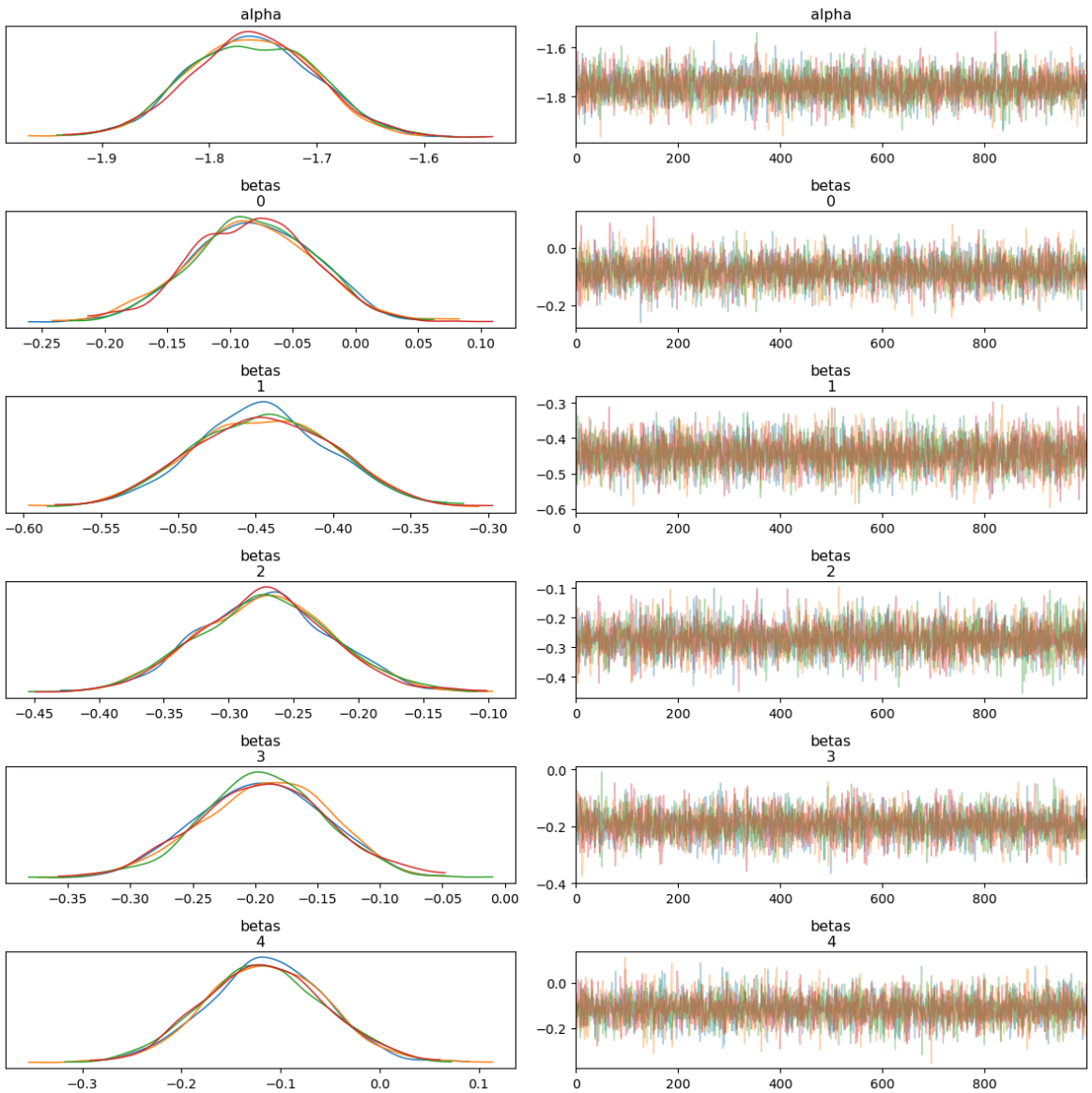
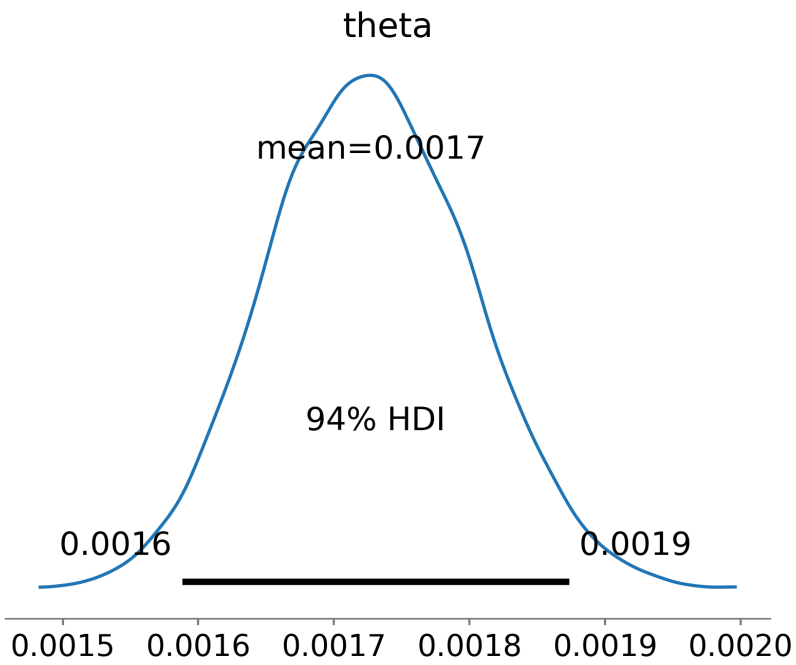


Figure 9 shows the MCMC traces for 4-chains for alpha and betas 0-4.

Convergence was confirmed by the **Gelman-Rubin** statistic $\hat{R}$ = ***1.0*** for all parameters, indicating that all four chains had converged to the same posterior distribution. Effective Sample Sizes (ESS) were **above 3,000** for all parameters, ensuring reliable posterior estimates.

The posterior distribution of θ from the global Binomial model is shown below, with a 94% Highest Density Interval (HDI) of [0.00168, 0.00177].



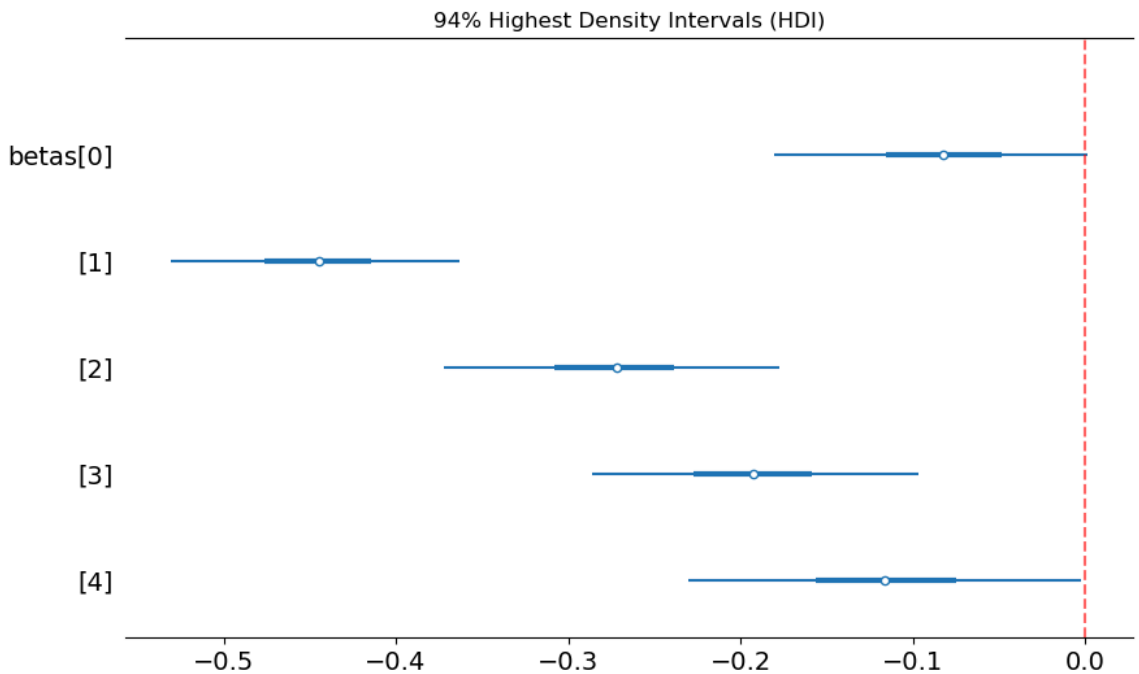


Figure 10 and 11 show the Posterior Distribution of θ from global Binomial distribution.

3.4: Threshold Optimisation (Objective 1)

Using the posterior mean coefficient estimates, a fraud probability was computed for each transaction in the test set. The Precision-Recall curve was then evaluated across all thresholds, and the threshold maximising the F1-score was selected.

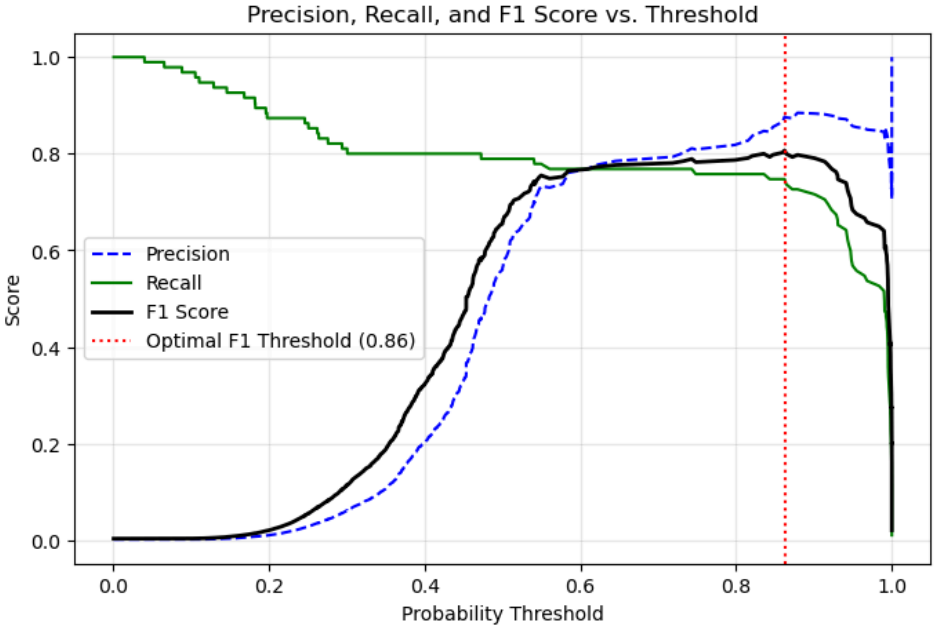


Figure 12 shows the Precision-Recall curve using Score vs. Probability Threshold graph.

3.5: Expected Loss Minimisation via Gradient Descent (Objective 2)

The expected loss function $L(\tau)$ was evaluated across a grid of thresholds. Gradient descent was applied to find the minimum, subject to a floor constraint preventing the threshold from dropping below a meaningful level. Without this constraint, the optimiser would lower the threshold for all large transactions, flagging a lot of normal transactions consequently.

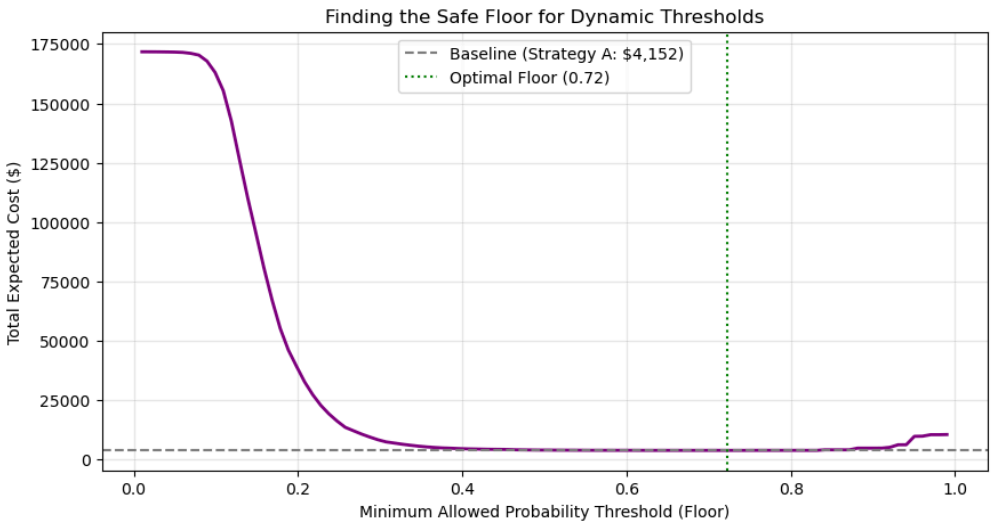


Figure 13 shows the baseline curve in total expected cost vs. min allowed prob threshold graph.

3.6: Confidence Interval for Daily Fraud Volume (Objective 3)

The confidence interval was computed using the posterior fraud rate $\hat{\theta} = 0.00172$ and the average daily transaction volume of 142,403. The 95% interval was constructed using the Normal approximation.

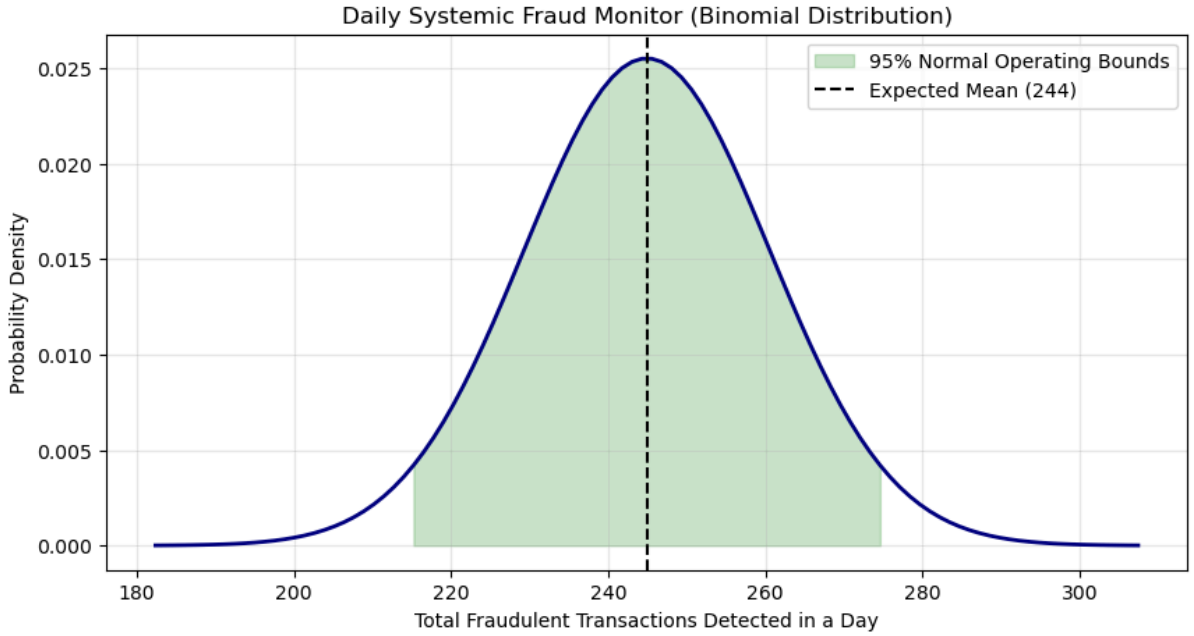


Figure 14 shows the Confidence Interval of daily frauds via Probability Density Graph.

4. Interpretation of Results

4.1: Objective 1: Probability, Prediction & Threshold of Frauds (Static Threshold)

The Bayesian model, when applied to the unseen test vault of 56,746 transactions, achieved:

Metric	Value
Optimal Threshold (max F1)	0.8623
Maximum F1-Score	0.8068
True Positives (Fraud caught)	71 out of 95
False Negatives (Fraud missed)	24
False Positives (Customers inconvenienced)	10
True Negatives (Normal cleared)	56,641

Table 4 shows the model result with normal threshold.

In plain terms: When the model's fraud probability score for a transaction exceeds 86.23%, it raises an alarm. At this setting, it correctly identifies 71 of the 95 real frauds (74.7% catch rate) while only inconveniencing 10 legitimate customers out of 56,651 (99.98% legitimate pass-through). This represents a carefully balanced trade-off.

The model was also assessed at the extremes: to catch 100% of frauds (all 95), the threshold would need to drop to just 4.04% - at which point, 56,315 out of 56,651 normal customers would be wrongly flagged. This illustrates why choosing the right threshold is a deliberate business decision, not a purely statistical one.

Compared to baselines, the Bayesian model's F1-score of 0.8068 is competitive with Random Forest (F1 = 0.83), however this observation leaves out two critical details. First, the training size for Random forest was 226,980, while MCMC model was trained on 1,890 data. Second, the distribution of both training set and test set was the same for RF, while MCMC had to adapt to a completely new real world distribution (test set). Additionally, Random Forest doesn't provide a fully probabilistic fraud score with quantified uncertainty.

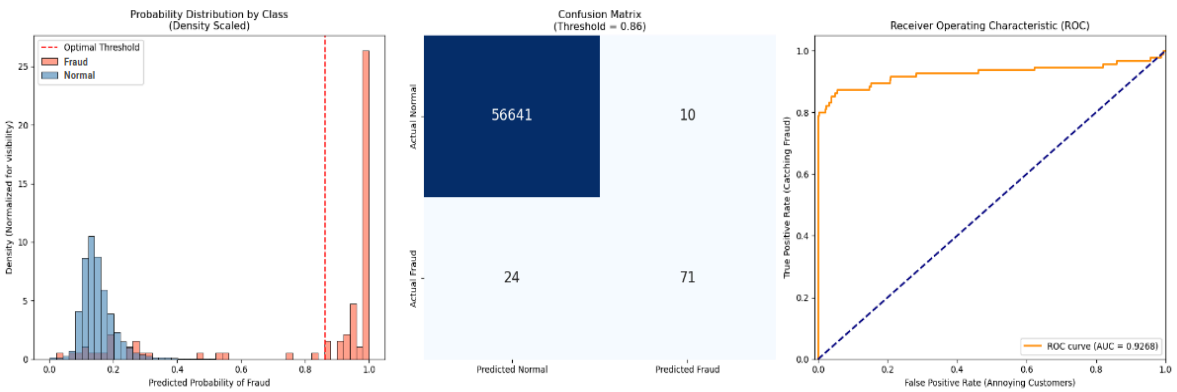


Figure 15, 16 and 17 show the Optimal Threshold, Confusion matrix and ROC curve of Bayesian Model.

4.2: Objective 2: Further Mitigation of Financial Loss

Let Strategy A be our original Bayesian model with fixed threshold 0.8623 and Strategy B be the dynamic threshold with floor at 0.7272.

The total fraud attempted in the test vault was €14,766.31. Under two strategies:

Strategy	Description	Total Financial Damage
Strategy A (Static Threshold = 0.8623)	Fixed threshold from Objective 1	€4,151.53
Strategy B (Dynamic Floor Threshold = 0.7227)	Gradient descent optimised	€3,913.24
Net Saving	B vs. A	€238.29

Table 5 shows the model comparison with static threshold vs dynamic threshold.

In plain terms: By dynamically adjusting the alarm threshold based on transaction amount - using a floor of 72.27% - the system saves an additional €238.29 compared to the static threshold, while recovering €10,853.07 of the €14,766.31 in attempted fraud. The key insight is that a €500 transaction with 40% fraud probability is more dangerous financially than a €1 transaction with 85% probability, so the threshold should respond to both probability and amount. This demonstrates that the model with higher accuracy (in this case F1-score) may not necessarily be the better model pertaining to financial interests of an institution.

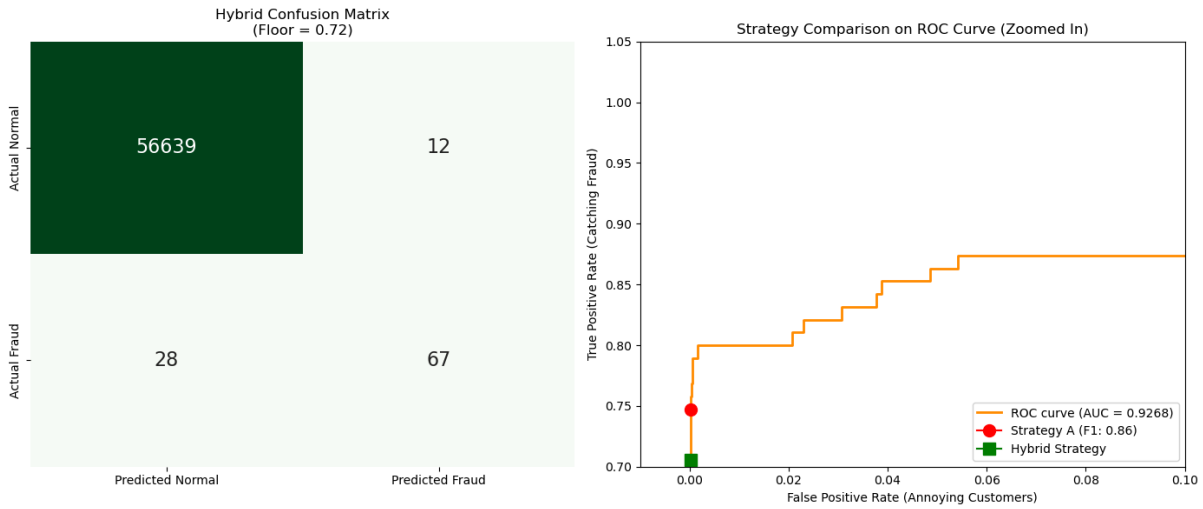


Figure 18 shows the Confusion matrix and the ROC curve of the model with Dynamic Floor Threshold.

4.3: Objective 3: Confidence Interval for Daily Fraud Counts

Based on the posterior fraud rate estimate and the average daily transaction volume:

Metric	Value
Average Daily Transactions	142,403
True Base Fraud Rate	0.172%
Expected Frauds Per Day	244
95% Safe Systemic Range	214 to 275 frauds per day

Table 6 shows the range of daily frauds in real world scenarios.

On a typical day, a bank should expect between 214 and 275 fraudulent transactions, given this transaction volume. If reported frauds consistently exceed 275 per day, it may indicate a coordinated attack or a new fraud vector emerging. Conversely, fewer than 214 per day may indicate that victims are not reporting fraud. This range gives risk management teams a data-driven benchmark for operational monitoring.

5. Critique

5.1: Model Performance Satisfaction

Objective 1: The model achieves a strong F1-score of 0.8068 and provides well-calibrated probabilistic predictions. It catches only 74.7% of frauds at the optimal threshold, meaning 25.3% of real frauds slip through. This is a massive improvement over not using a model in the first place, and performs marginally better than Baseline. The model provides the critical advantage of knowing how uncertain it is about each prediction which is something that no standard machine learning (ML) classifier offers.

Objective 2: The gradient descent approach correctly identifies that a dynamic threshold sensitive to transaction amount is financially superior to a static one. The €238.29 net saving in a 56,746-transaction vault, while modest, scales linearly with transaction volume.

Objective 3: The Poisson-based confidence interval provides a practical, interpretable range for systemic fraud monitoring. The derivation is mathematically sound and the assumptions are reasonable.

5.2: Limitations

1. **Computational constraints:** The MCMC sampler was trained on only 1,890 transactions (0.66% of the full dataset) to keep runtime manageable. This severely limits the model's exposure to the diversity of fraud patterns, potentially degrading generalisation.
2. **Encrypted features:** Since V1–V28 are PCA-transformed with no available feature dictionary, it is impossible to interpret what specific transaction characteristics the model has learned to weight. This makes the model a "black box" in terms of business interpretability, despite its probabilistic framing.
3. **Static cost assumptions:** The €10 false-positive cost is an assumption, not a measured figure. Real friction costs (customer service calls, chargeback processing, reputational damage) are likely higher and variable. The model's financial conclusions are sensitive to this choice.
4. **No temporal dynamics:** The model treats all transactions identically regardless of when they occurred. When real fraud patterns evolve over time, fraudsters adapt their methods. A model trained on this two-day snapshot may become stale quickly.
5. **Label reliability:** Frauds are defined by customer reports. Silent frauds (transactions the customer did not notice or report) are labelled as "normal," biasing the model toward under-detecting low-amount frauds.

5.3: Future Work

1. **Sequential RF + MCMC pipeline:** Random Forest could be used as a first-pass filter to create a high-recall candidate set, and MCMC sampling could then focus only on borderline cases. This would dramatically reduce computational cost while maintaining coverage.
2. **Full dataset MCMC with GPU acceleration:** With access to a GPU-accelerated MCMC engine (e.g., NumPyro with JAX), the full 283,726-row dataset could be used for training, likely improving recall significantly.
3. **Hierarchical time-series model:** A dynamic Bayesian model that incorporates temporal drift in fraud patterns would better reflect real-world conditions, where fraudsters adapt over weeks and months.
4. **Cost calibration study:** Partnering with a financial institution to obtain actual false-positive operational costs would make the loss minimisation (objective 2) far more actionable.
5. **Multi-threshold personalisation:** Different customers may have different risk tolerances. A personalized threshold per user segment (high-value customers vs. standard accounts) could further reduce both fraud losses and customer friction.

6. Appendices

Appendix A: European Credit Card Transactions Dataset

The dataset used in this project is the publicly available **European Credit Card Transactions Dataset**, containing transactions made by European cardholders over a two-day period in September 2013. The dataset is available at the following link: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Appendix B: Posterior Summary for Global Fraud Rate (Theta)

The following table is the full ArviZ posterior summary for the Binomial model of global fraud rate, saved to posterior_theta_summary.csv:

Parameter	Mean	SD	HDI 3%	HDI 97%	MCSE Mean	MCSE SD	ESS Bulk	ESS Tail	$\hat{R}$
theta	0.002	0.0	0.002	0.002	0.0	0.0	3255	3180	1.0

The $\hat{R}$ = 1.0 confirms perfect convergence. ESS values above 3,000 indicate the sampler explored the posterior thoroughly.

Appendix C: Repository Structure

```
Credit-Card-Fraud-Detection/
├── project.ipynb                # Experimentation of dataset
├── project1.ipynb               # 4-chain MCMC run and test evaluation
├── colab.ipynb                  # Google Colab version
├── Model.py                     # Global Binomial MCMC model
├── Model.md                     # Mathematical model specification
├── databasein.py                # SQLite data ingestion script
├── dbcheck.py                   # Database integrity checker
├── db_tools_test.py             # Unit tests for database tools
├── test_harness.py              # Model testing harness
├── environment.yml              # Conda environment specification
├── creditdatabase.db            # SQLite database (cleaned transactions)
├── my_4chain_model.nc           # Saved ArviZ trace (NetCDF format)
├── posterior_theta_summary.csv  # Exported posterior statistics
├── bayes_tools/                 # Custom module: model building, DAGs, diagnostics
├── db_tools/                    # Custom module: database loaders
├── synth_tools/                 # Custom module: synthetic data tools
└── [*.png]                      # All generated visualisation files
```

Appendix D: Environment Specification

The project was developed in a Conda environment with the following core dependencies:

- **Channels:**
  - conda-forge
  - defaults
- **Dependencies:**
  - python=3.10
  - pymc>=5.0
  - arviz>=0.16
  - scikit-learn>=1.2
  - pandas>=2.0
  - numpy>=1.24
  - matplotlib>=3.7
  - seaborn>=0.12
  - scipy>=1.10
  - sqlite3
  - graphviz
  - causalgraphicalmodels

Appendix E: GitHub Repository

All code, data, and visualisations are available at:
https://github.com/Vaibhav9892/Credit-Card-Fraud-Detection